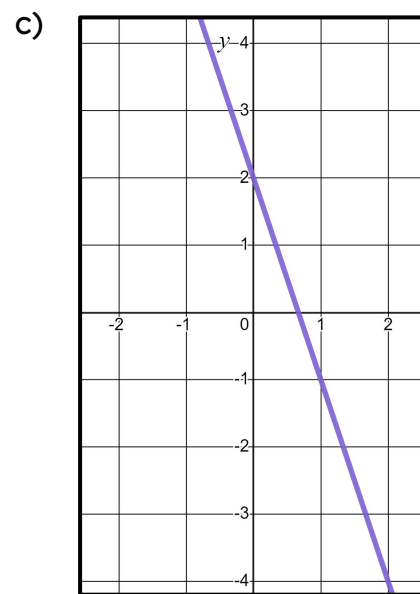
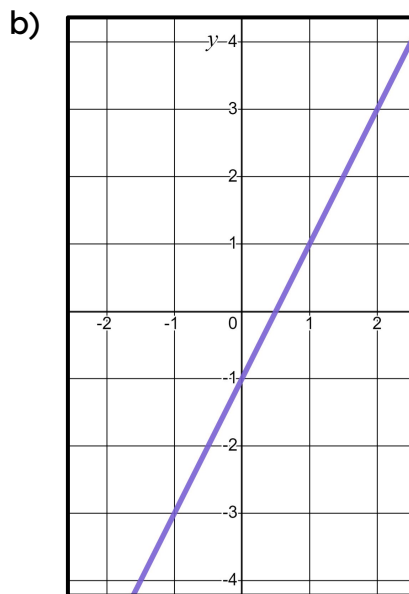
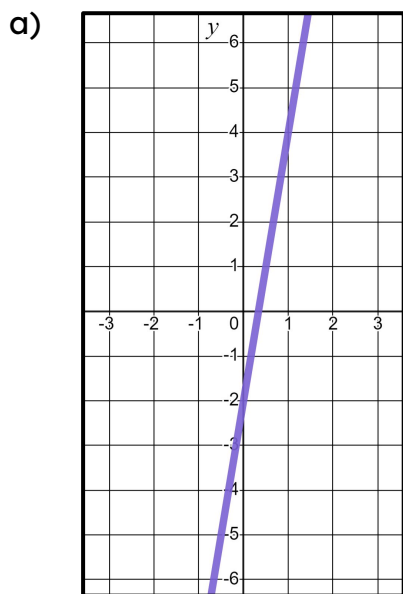




Equations and their graphs

Task A: Reviewing features of linear graphs

1) Gradient of 6 and a y -intercept of $(0, -2)$ is a description unique to graph (a). Write descriptions for b) and c)



b) Gradient of 2 and a y -intercept of $(0, -1)$

c) Gradient of -3 and a y -intercept of $(0, 2)$

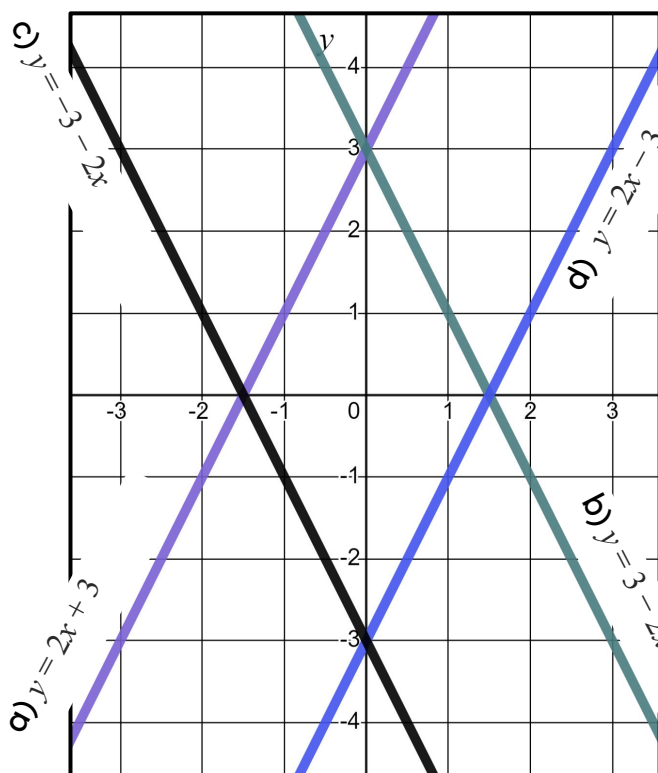
2) Match the linear equations to their respective lines.

a) $y = 2x + 3$

b) $y = 3 - 2x$

c) $y = -3 - 2x$

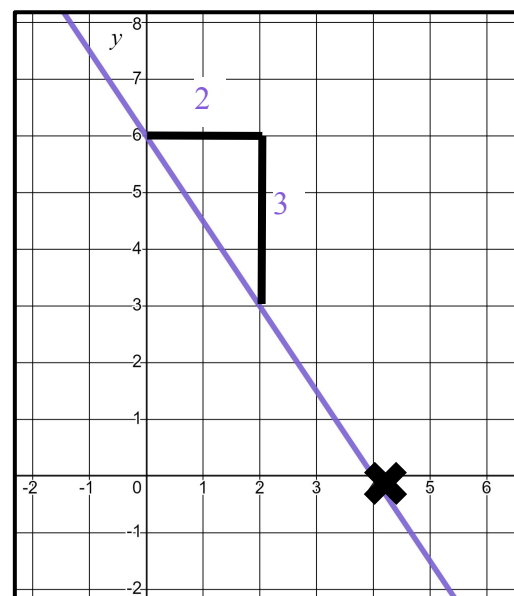
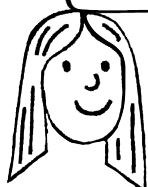
d) $y = 2x - 3$



3) Laura is writing the equation of this linear graph.

I see a change of 3 in y and a change of 2 in x and an intercept of 4 so this is the line

$$y = \frac{3}{2}x + 4$$



Write at least two sentences to explain Laura's errors to her. You may choose to annotate the graph in your explanation.

The gradient is the change in y divided by the change in x but with respect to the *positive* x -direction.

In the form $y = mx + c$ the c relates to the *y-intercept*, not that of the x axis.

This is the line $y = 6 - \frac{3}{2}x$

Task B: Sketching linear graphs

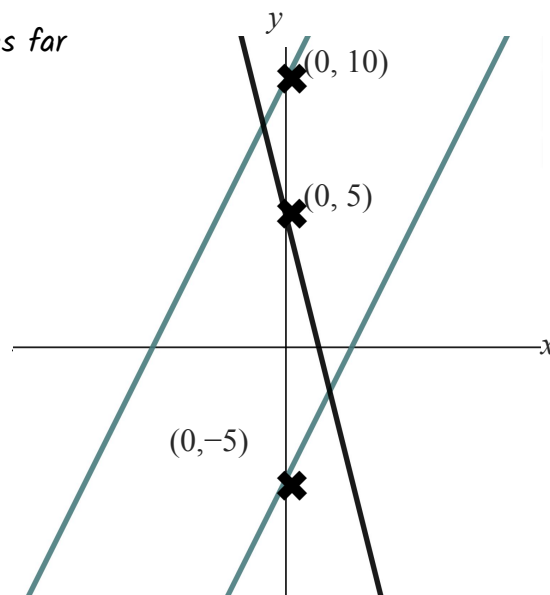
1) Sketch the lines $y = 2x + 10$, $y = 5 - 4x$, and $y = 2x - 5$ on the below axes.

y-intercepts should be labelled and $(0, 10)$ should be twice as far from the origin as $(0, 5)$

$(0, -5)$ should be the same distance from the origin as $(0, 5)$

$y = 2x + 10$ and $y = 2x - 5$ should have the same gradient therefore look parallel in your sketch.

$y = 5 - 4x$ is a steeper gradient. This should be visibly obvious.



2) Match the linear equations to their respective lines.

a) $y = 2x - \frac{1}{4}$

b) $y = 18 - 2x$

c) $y = -5x + 18$

d) $y = \frac{1}{4}x - 2$

